# MILITARY SPECIFICATION

## FILTER ELEMENT, FLUID, PRESSURE, FFU-3/E

### 1. SCOPE

1.1 Scope. This specification covers one type of pressure fluid filter element, designated FFU-3/E.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| L-P-378 | Plastic Film (Polyethylene Thin Gage) |
| PPP-B-636 | Box, Fiberboard |
| PPP-B-640 | Box, Fiberboard, Corrugated, Triple-Wall |
| PPP-T-60 | Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-B-121 | Barrier Material, Greaseproofed, Waterproofed, Flexible |
| MIL-G-5572 | Gasoline, Aviation, Grades 80/87, 100/130, 115/145 |
| MIL-T-5624 | Turbine Fuel, Aviation, Grades JP-4 and JP-5 |
| MIL-F-8901 | Filter-Separators, Aviation and Motor Fuel, Ground and Shipboard Use, General Requirements and Test Procedures for |
| MIL-P-19834 | Plates, Identification, Metal Foil, Adhesive Backed |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of U. S. Military Property |
| MIL-STD-831 | Test Reports, Preparation of |
| MIL-STD-882 | Systems Safety Program, for Systems and Associated Subsystems and Equipment, Requirements for |
| MS29513 | Packing, Preformed, Hydrocarbon Fuel Resisting, O-Ring |



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(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing Materials

| | |
|---|---|
| D256 | Standard Methods of Test for Impact Resistance of Plastics and Electrical Insulating Materials |
| D638 | Method of Test for Tensile Properties of Plastics |
| D790 | Standard Method of Test for Flexural Properties of Plastics |

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General specification. The requirements of MIL-F-8901 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Materials.

3.3.1 Metallic parts. All metallic parts of the filter element shall be fabricated from an aluminum alloy suitable for the application.

3.3.2 Fungusproof materials. Materials that are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent.

3.3.3 Petroleum products. The materials used in the construction of the filter element shall neither affect nor be affected by petroleum products.

3.3.4 Plastic parts. If used in the construction of the filter elements, molded plastic parts shall have the following physical properties:

a. Tensile strength, minimum - 4,500 psi
b. Flexural strength, minimum - 7,500 psi
c. IZOD impact, minimum - 1.5 pounds per inch of notch

3.4 Design and construction. The filter element shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and flight line service.

3.4.1 Functional design. The filter element shall be designed for installation in Bowser-Briggs Filtration Company models BFS-50-600 and BFS-20-300 filter-separator conversion kits. The filter element shall be designed as shown on figure 1 and shall be adaptable to the element canisters and element



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seal plate shown on figures 2 and 3. The conversion kit and pressure vessel furnished for preproduction testing shall be modified by the element supplier to provide a higher strength element mounting plate and second stage, where this is necessary to withstand the tests required herein and in MIL-F-8901. The second stage shall consist of either a 100- or 200-mesh tetrafluoroethylene -coated screen(s) of suitable area for the application and installation. The element shall be provided with an aluminum alloy center tube.

3.4.1.1 Compatible fuels. The filter element shall be so designed that it can be used with fuels conforming to MIL-G-5572 and MIL-T-5624.

3.4.1.2 Canister gaskets. Unless otherwise specified (see 6.2), a gasket shall be furnished with each element for installation between the canister assembly and canister end seal plate (see figures 2 and 3). The material for the gasket shall be compatible with fuels conforming to MIL-T-5624 and shall be of a thickness consistent with design.

3.5 Performance. The filter elements shall meet the performance requirements of MIL-F-8901 when the indicated number are installed in a filter-separator vessel which has been converted by the addition of a BFS-26-300 or BFS-50-600, as applicable, conversion kits:

a. 25 elements - Model BFS-50-600 at a flow rate of 600 GPM
b. 13 elements - Model BFS-26-300 at a flow rate of 300 GPM

3.6 Identification of product. The following information shall be marked on each filter element and each separator bag in accordance with MIL-STD-130. If the item is marked on any sealing surface, the markings shall not affect the sealing of the element.

Manufacturer's part number_____
Manufacturer's name or trademark_____
Date of manufacture_____

3.6.1 Manufacturer's part number. The manufacturer's part number and drawing number shall be the same.

3.6.2 Nameplate. A nameplate fabricated of type I materials in accordance with MIL-P-19834 shall be supplied, one per each 13 elements to be shipped. The nameplate is intended to be attached by Government personnel to the pressure vessel to facilitate maintenance of the fueling system. The nameplate shall conform to figure 4. Attaching instructions shall be included with the name-plate. When specified (see 6.2), additional nameplates shall be provided.

3.7 Government loaned property. When specified in the contract (see 6.2), the Government will loan the filter-separator vessel to the contractor upon his request.

3.8 Workmanship. The filter element, including all parts, shall be construc-ted and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of marking of parts and assemblies and freedom of parts from burrs and sharp edges.

3.9 Safety. All equipment under this specification shall be analysed for category III and category IV hazards as defined in MIL-STD-882. Any hazard so



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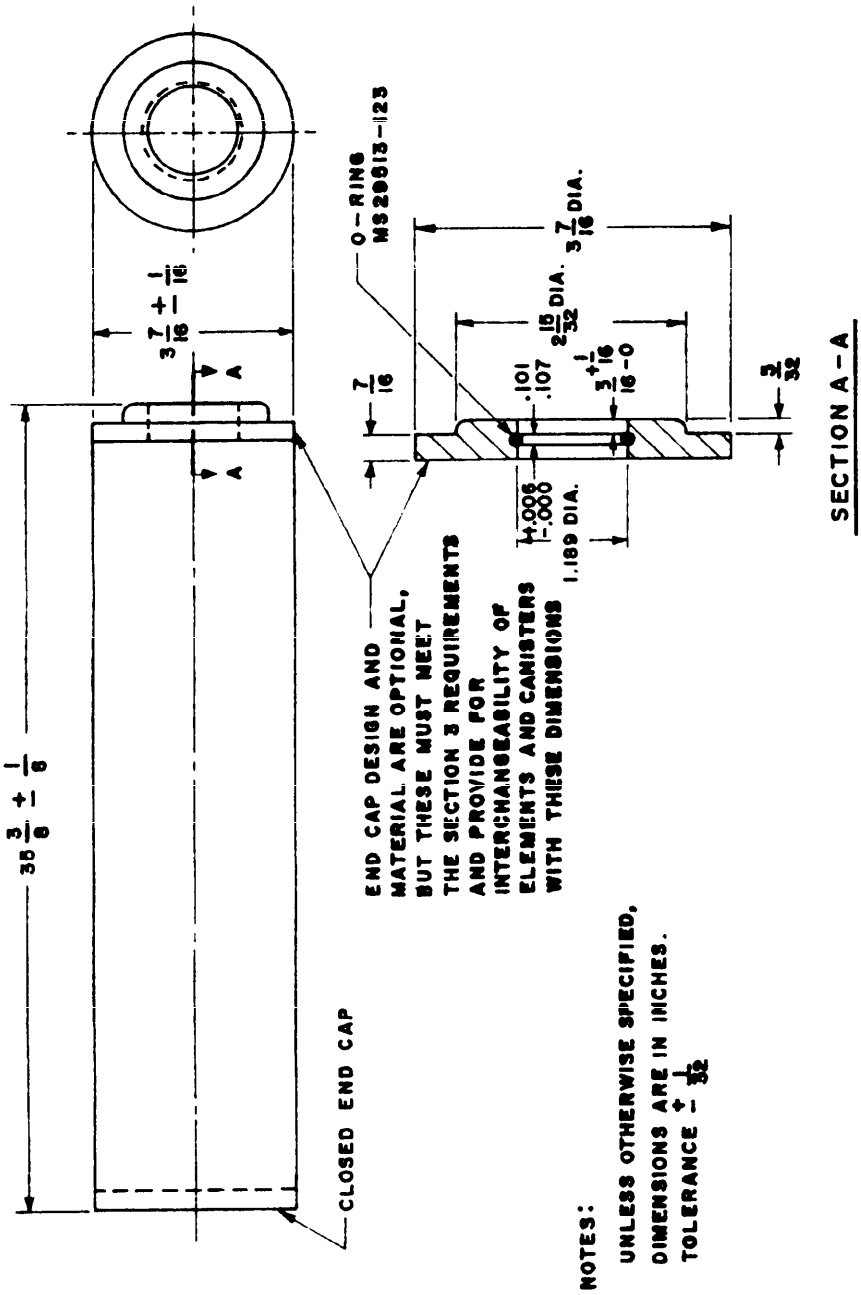



FIGURE 1. FILTER SEPARATOR ELEMENT



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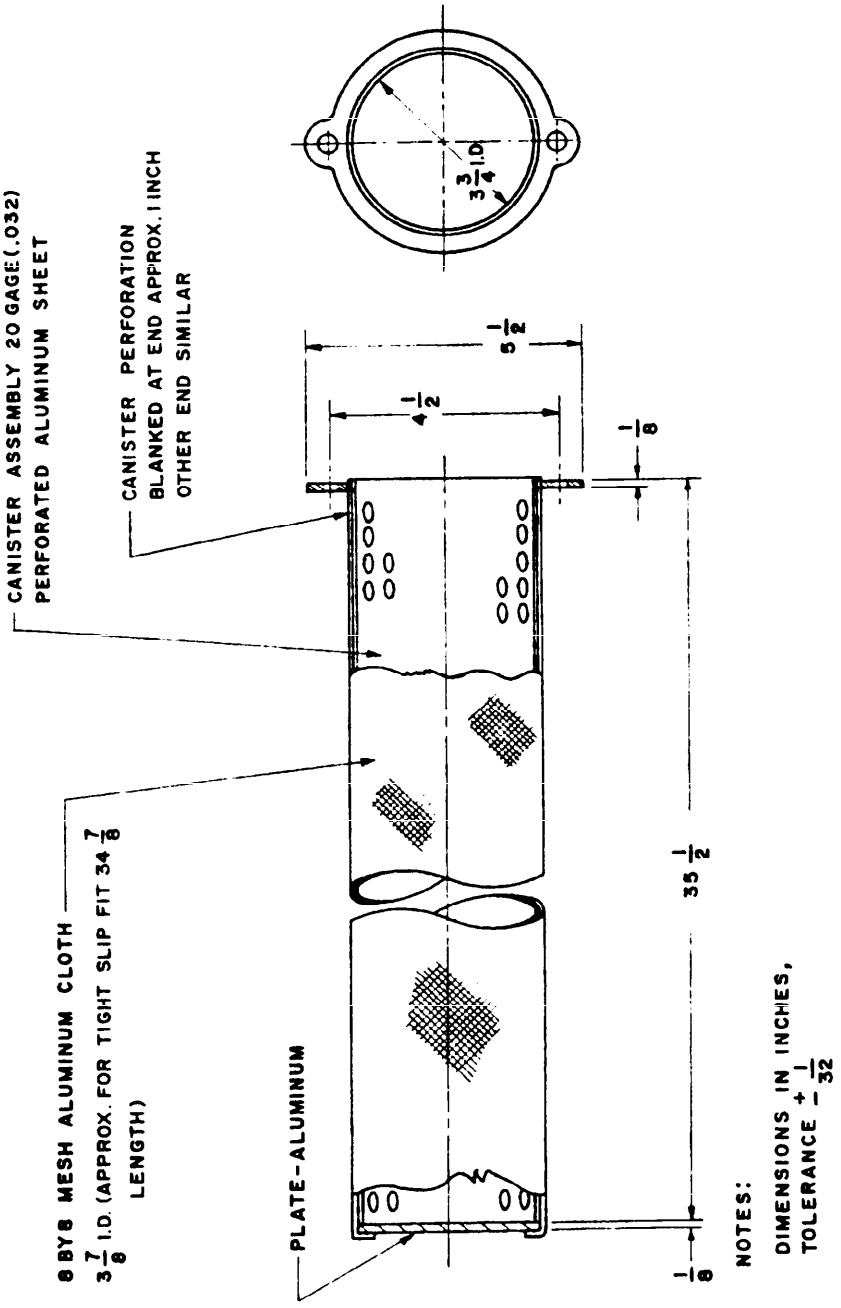
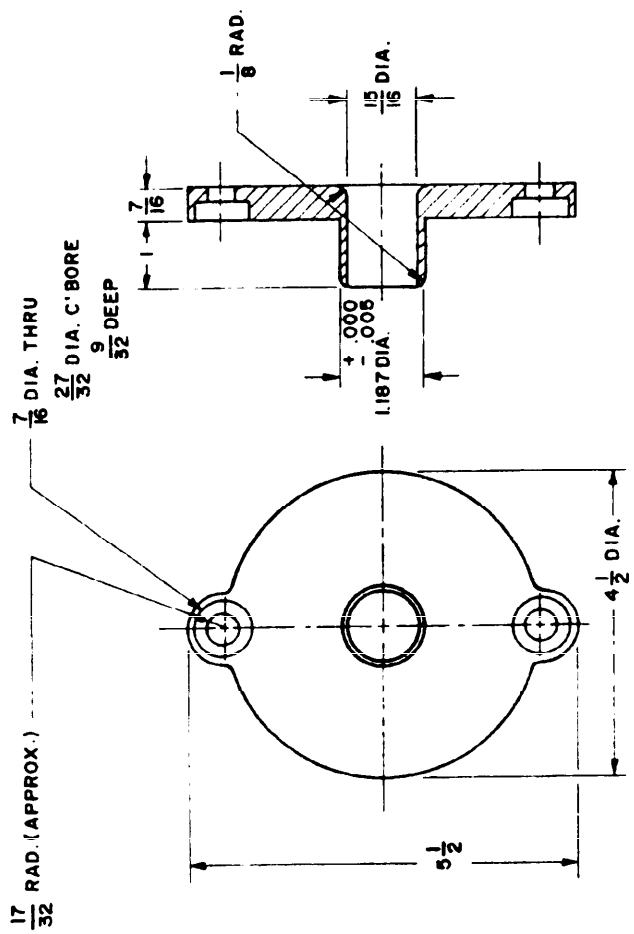



FIGURE 2. CANISTER ASSEMBLY



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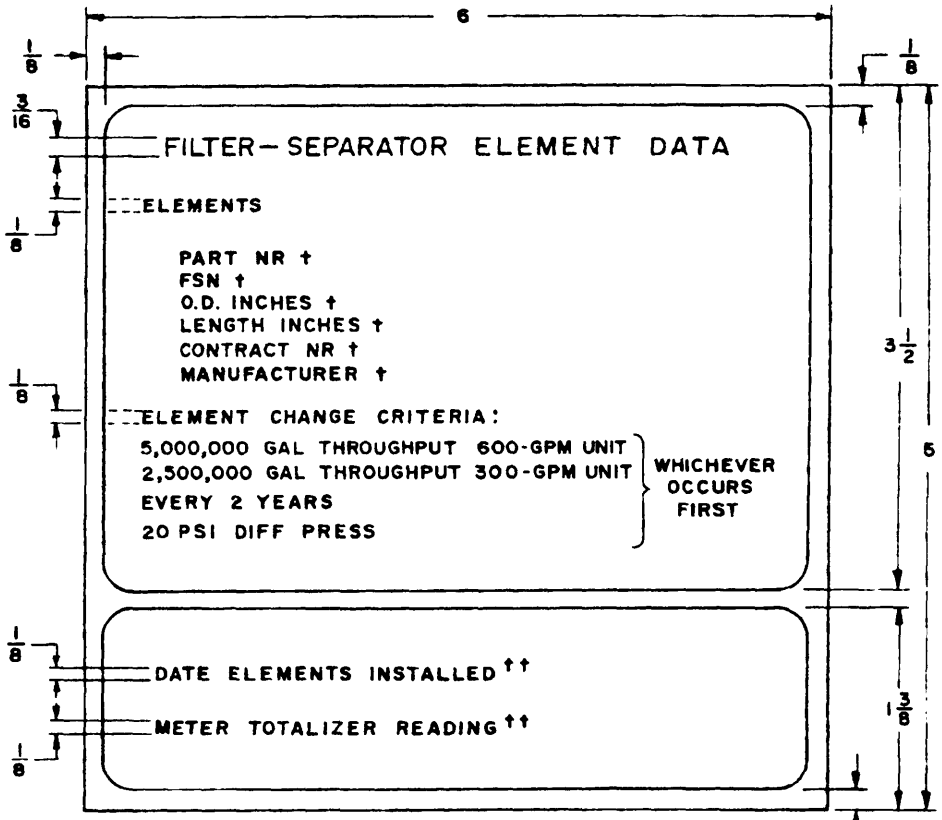



NOTES:
MATERIAL: ALUMINUM (6061)
DIMENSIONS IN INCHES
TOLERANCE ± 1/32

FIGURE 3. CANISTER END SEAL PLATE



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NOTES:

1. ALL LETTERING SHALL BE $\frac{3}{32}$-INCH HIGH, UNLESS OTHERWISE NOTED
2. BACKGROUND SHALL BE BLUE
3. DIMENSIONS IN INCHES, TOLERANCE $\pm \frac{1}{32}$

FIGURE 4. NAMEPLATE



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identified shall be brought to the attention of the procuring agency with recommended action for satisfying safety requirements. Action order of precedence shall be as specified in MIL-STD-882, para 5.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection and testing. The inspection and testing of the filter element shall be conducted in a full-scale filter-separator unit in accordance with MIL-F-8901 and as specified herein.

4.2 Test conditions. As specified (see 6.2), full scale testing shall be performed in either of the following Bowser-Briggs Filtration Company filter-separator conversion kits with the number of elements installed as indicated:

a. Model BFS-50-600 - 25 elements
b. Model BFS-26-300 - 13 elements

4.3 Preproduction testing (see 6.2).

4.3.1 Test samples. When the Bowser-Briggs Filtration Company model BFS-50-600 filter-separator conversion kit is furnished (see 6.2), the preproduction test samples shall consist of 101 elements and 25 canister gaskets. When the Bowser-Briggs Filtration Company model BFS-26-300 filter-separator conversion kit is furnished, the test samples shall consist of 53 elements and 13 canister gaskets. The samples shall be identified with the manufacturer's part number and such other information as is required by the procuring activity. Samples shall be prepared as follows:

4.3.1.1 Data to accompany test samples. The preproduction test samples shall be accompanied by the following data:

a. Brief operating and installation data to enable test personnel to correctly install and operate the equipment.

b. Engineering data in the form of assembly drawings (two sets).

4.3.1.2 Test sample and data for procuring activity. When tests are conducted at a site other than the laboratory of the procuring activity, the following shall be furnished that activity:

a. Three copies of a test report in accordance with MIL-STD-831. In addition to results, the report shall contain complete records of the tests, including data sheets, performance curves, chronological test records, photographs, sample calculations, test procedures, and a description of the test apparatus.



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b. A sample element from the filter-separator which was subjected to the preproduction tests.

c. Color photograph of the sample element.

d. A new filter element.

4.3.2 Preproduction tests. The preproduction tests shall consist of all tests specified in MIL-F-8901, except for the following changes:

a. The water removal tests shall be performed at the fuel flow rate indicated for the conversion kits as noted in 4.5.3. During the second hour, water shall be introduced at a rate of 3 percent by volume in lieu of 5 percent by volume. Water removal tests at 10 percent by volume during the third hour are not required.

b. During the inhibited fuel test, a single-pass fuel transfer system is required. Tests of filter-coalescer-separator elements or media combinations devised to represent full-scale equipment shall be designed for testing at flow rates of not less than 23 gpm per element.

c. Upon completion of the life test specified in MIL-F-8901, the filter-separator housing shall be disassembled and inspected to determine the condition of the filter elements. Defects in the filter element such as swelling of the element, damaged elements, damaged gaskets, et cetera, shall be recorded. Excessive swelling of or damage to the element or other parts shall be cause for rejection.

d. Sections of plastic material used in the construction of the element and caps or especially prepared test specimens shall be tested as indicated below to verify plastic material compliance. This test data developed shall be included in the test report:

(1) Tensile strength - ASTM D638, type I
(2) Flexural strength - ASTM D790
(3) IZOD impact strength (notched) - ASTM D256, method A

4.4 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests See 4.4.1
b. Sampling test See 4.4.2

4.4.1 Individual tests. Each filter element shall be subjected to the test specified in para 4.5.1.

4.4.2 Sampling plan and test. Samples of the filter element shall be selected at random from each lot in accordance with MIL-STD-105 and subjected to the test specified in 4.5.2. An acceptable quality level (AQL) of 1.5 percent defective, level I, shall be established. The test shall be considered destructive.

4.5 Test methods.

4.5.1 Examination of product. The filter element shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.



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4.5.2 Filter-coalescer element tests. The filter element shall be subjected to the test of MIL-F-8901 entitled, Filter-coalescer element.

4.5.3 Fuel flow rate. Under the preproduction test conditions, the fuel flow rate shall be as follows for Bowser-Briggs Filtration Company filter-separator conversion kits indicated:

a. 600 gpm - Model BFS-50-600
b. 300 gpm - Model BFS-26-300

4.6 Inspection of preparation for delivery. Preservation, packaging, packing and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Cleaning and drying. Prior to packaging, filter elements and attaching hardware, if any, shall be thoroughly cleaned and dried in accordance with the applicable processes of MIL-P-116.

5.1.2 Level A. After cleaning and drying, filter elements and attaching hardware, if any, shall be packaged, one each, in accordance with method IC-1 of MIL-P-116, modified as follows:

5.1.2.1 Packaging changes. Each filter element shall be inspected and sealed in a polyethylene bag or film conforming to L-P-378 and placed in a fiberboard container conforming to PPP-B-636, grade W5C. Bags and boxes shall be of minimum size consistent with the size of filter elements being packaged. Containers shall be closed and sealed in accordance with the applicable specification and appendix thereto.

5.1.2.2 Nameplates. Each nameplate (see 3.6.2) shall be packaged in accordance with method III of MIL-P-116 by wrapping in grade A barrier material conforming to MIL-B-121 and securing with tape conforming to PPP-T-60.

5.1.3 Level C. Preservation and packaging shall be the minimum necessary to afford protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing (see 6.2).

5.2.1 Level A. Filter elements shall be packed in overseas-type containers conforming to PPP-B-640. Unless otherwise specified, the nameplates of 3.6.2 shall be included in the individual containers in quantities of one for each 13 filters or fraction thereof (see 6.2). Containers shall be of minimum weight and cube, of uniform size and shape, and shall contain like quantities, when practical. Closure of the containers shall be in accordance with the applicable specification and appendix thereto.

5.2.2 Level B. The filter element shall be packed as specified in 5.2.1, except containers shall be of the domestic type and closure shall be in accordance with the requirements for domestic containers.



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5.2.3 Level C. Packages which require overpacking for acceptance by the carrier shall be packed in shipping containers in a manner that will insure safe delivery at the lowest transportation and packing charges. Unless otherwise specified, the nameplates of 3.6.2 shall be included in the individual containers in quantities of one for each 13 filters or fraction thereof (see 6.2). Containers shall meet consolidated freight rules and regulations or other rules and regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The FFU-3/E filter element is intended for use as a replacement item in ground aviation liquid fuel filtering and servicing systems for the Bowser-Briggs Company models BFS-50-600 and BFS-26-300 filter-separator conversion kits.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. If canister gaskets are not to be supplied (see 3.4.1.2).

c. The model (BFS-50-600 or BFS-26-300) Bowser-Briggs Filtration Company filter-separator conversion kit that will be specified during the preproduction tests (see 4.2).

d. Location and conditions for preproduction testing (see 4.3).

e. The quantity of additional nameplates required (see 3.6.2) and shipping instructions for them (see 5.2.1 and 5.2.3).

f. Applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

g. Special shipment marking (see 5.3).

h. If Government-loaned filter-separator vessel is available to contractor (see 3.7).

6.3 Definition. For the purpose of this specification, the following definitions will apply.

6.3.1 Lot. All filter elements of the same part number, manufactured in a production run and submitted for acceptance at the same time shall be considered a lot for acceptance purposes.

6.4 Contract administration.

6.4.1 Test system. The contracting organization should require that a prospective contractor shall have, or can contract for, a suitable test system prior to the award of any contracts for filtration equipment. A test system is the piping, test requirement apparatus, arrangement of test components, laboratory equipment, laboratory supplies, and techniques used during the evaluation of fuel filtration equipment.



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6.4.2 Government inspection. The contracting organization should provide for Government inspection during the preproduction buildup and test in addition to providing Government inspection after production is released.

6.4.3 Latest issue of document. The organization should, prior to issuance of a procurement bid or proposal (IFB/RFP), contact the engineering activity having technical responsibility for the specification to confirm that the latest edition of the specification is being cited.

6.5 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian
Air Force - 84

Preparing activity:
Air Force - 84

Reviewer:
Air Force - 11

Project number:
4330-F138



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